Class-Specific Errors and Deferral under Molecular Distribution Shift: A Budget-Controlled Comparison of Conformal and Abstention Rules

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Abstract

Prediction sets for shifted molecular assays can attain higher empirical coverage by expanding more sets. Evaluating a method only by its coverage gain therefore leaves an important question unanswered: does it identify erroneous decisions more effectively than a simpler rule with the same deferral budget? We formulate this question for binary assay activity, distinguishing false-inactive decisions, false-active decisions, and requests for further evidence. The revised study compares marginal and class-conditional conformal prediction, estimated density-ratio weighting, and decision-specific probability, uncertainty, similarity, and random deferral using common molecular predictions and matched output budgets. The previously proposed transport-audited multi-view envelope (TAME) is evaluated as an empirical set-expansion rule, alongside its single-view, ungated, and protected-label variants. Calibration uses the finite-sample order statistic with an infinite threshold when the requested rank is unavailable. The analysis separates historical Tox21 development and EPA findings from the new retrospective comparisons, preserves repeated predictions for each molecule in paired inference, and reports assay-specific error–deferral trade-offs. A historical comparison with ordinary stacked

Mondrian prediction found a 1.36-percentage-point improvement in mean minimum-class coverage at a 1.61-point loss in macro correct-singleton yield; a simple equal-logit comparator had slightly better values on both quantities in the descriptive twelve-assay summary. These observations motivate the revised comparisons rather than establish TAME superiority. The revised experimental results remain pending and will determine whether transport information improves error targeting beyond matched deferral.

Keywords: molecular assay prediction; conformal prediction; class-specific error; selective prediction; covariate shift; deferral

1 Introduction

Molecular prediction systems often supply a probability of assay activity for compounds that differ from their training collection. A screening workflow may use that probability to designate a compound as inactive, flag it as active, or request additional evidence. These actions have different consequences. Incorrectly designating a positive compound as negative and a negative compound as positive should therefore be counted separately, alongside the number of compounds deferred. A method that produces fewer errors by making almost no decisions may have little practical value.

Conformal prediction provides a principled way to construct prediction sets under specified sampling assumptions. Its standard marginal guarantee does not imply equal coverage within every outcome class. Class-conditional Mondrian calibration addresses the outcome-class distinction under exchangeability within the corresponding classes [1]. Neither construction automatically transfers that guarantee to arbitrary molecular distribution shift. These are established properties, rather than new principles derived from the present molecular examples.

Under covariate shift, likelihood-ratio weighting relates source and target calibration distributions [2]. Molecular implementations must estimate those ratios. CoDrug combines a trained prediction model with density estimation for molecular conformal prediction [3]; KMM-CP uses selective kernel mean matching for covariate-shift correction [4]. Such methods are directly relevant comparators. Their representation learning, ratio estimation, support selection, and prediction-set construction should be distinguished from a generic domain-classifier weighting baseline.

An alternative intervention is to expand an existing prediction set. In binary classification, adding a

missing label to a singleton converts a decision into a deferral. Coverage cannot decrease on the same labeled observations, but this is a consequence of set inclusion. It does not establish that the selected compounds were better targets for deferral than those chosen by probability uncertainty, chemical novelty, or random selection. Nor does it establish a portable rule for selecting a protected class.

We examine whether a two-view transport-based expansion rule, TAME, targets class-specific decision errors more effectively than a simpler probability rule at the same deferral budget. Matching is performed separately within the two original singleton classes, so neither the number nor the original class composition of deferred decisions can explain an advantage. We also identify the restricted setting in which constant query weights make this expansion exactly equivalent to probability ranking. Query-weight alignment controls, component ablations, and controlled shifts then test the possible source and limits of additional targeting information. This is an empirical and structural comparison, without a new conformal-validity theorem.

The evaluation covers twelve Tox21 assays and four additional endpoint families, with shared-predictor comparisons and complete paper-based CoDrug and KMM-CP reference systems. The endpoint meanings remain separate. In particular, Tox21 activity denotes a nuclear-receptor or stress-response assay measurement; it is not interchangeable with clinical toxicity [6]. Fixed molecular cohorts, class-specific decision metrics, and molecule-cluster inference make the error–deferral question explicit across these settings.

2 Methods

2.1 Data roles and scope of inference

The Tox21 10K collection supplies labeled training data. The public leaderboard collection contains 296 structures, and the released final EPA collection contains 647 structures before standardization and overlap exclusions. All twelve official assays are retained. The six assays used in the original primary panel—NR-AhR, NR-ER, SR-ARE, SR-ATAD5, SR-MMP, and SR-p53—remain an explicitly historical panel selected by at least twenty active labels in the public development collection. Results for this panel and all twelve assays are reported separately.

The EPA labels were accessed during the original study. Every comparison newly designed after peer review is therefore retrospective, including reruns with different random seeds. These analyses measure a

fixed retrospective comparison on the observed collection; new random seeds do not undo prior use of its outcomes in method development.

The original source files, standardization rules, and exclusions remain versioned. Reconstruction uses the original cleaning environment's RDKit version, 2026.03.4. Invalid structures receive a reason code; missing assay outcomes are excluded only from that assay's labeled evaluation. Canonical structure identity determines overlap checks and connects repeated observations for statistical analysis. Each method uses the same eligible rows within an assay, model, and seed. Counts before and after every exclusion must accompany the results.

2.2 Common predictions and calibration roles

The controlled comparison uses 2048-bit radius-two Morgan fingerprints. The historical 60/20/20 allocation separates model fitting, stack fitting, and conformal calibration. A logistic predictor and the original four-component stack provide base-predictor comparisons. The stack consists of logistic regression, random forest, extra trees, and Bernoulli naive Bayes; its second-stage logistic model is trained only on the stack-fitting role. The equal-logit combination is retained because the archived results identify it as a relevant simple comparator. A method comparison never changes the underlying probability estimates unless it is explicitly labeled an end-to-end model comparison.

Models, data roles, calibration rows, and target rows are identical across the uncertainty layers being compared. Least-ambiguous-classification (LAC) scores are the primary scoring rule. A deterministic adaptive-prediction-set (APS) score provides a scoring sensitivity analysis [5]. Predictor and score effects are reported separately, rather than allowing either to be confused with a gain from transport weighting.

2.3 Additional assay families and fixed outer splits

The extension evaluates the AMES, standard hERG, DILI, and CYP2D6_Veith benchmarks distributed through Therapeutics Data Commons [7–10]. Positive outcomes mean Ames mutagenicity, hERG blockade, the liver-injury-associated class, and CYP2D6 inhibition-positive in the respective source benchmark. The CYP2D6 assay signal can arise from an inhibitor or a competing substrate; its binary label does not establish clinical toxicity or a unique mechanism. These outcomes are reported separately. The hERG task is the standard small collection, not hERG Central or the Karim variant; label rules from those other collections

are not imported.

Structures are processed with the same RDKit 2026.03.4 standardizer as the Tox21 reconstruction. Same-label duplicate structures are collapsed deterministically, and all rows of a structure group with conflicting binary outcomes are removed. The resulting source tables contain 7,255 AMES structures (3,304 negative and 3,951 positive), 607 hERG structures (193 negative and 414 positive), 474 DILI structures (238 negative and 236 positive), and 13,083 CYP2D6 structures (10,583 negative and 2,500 positive). These counts describe the fixed inputs and do not represent model-performance findings.

Each endpoint has a fixed random outer split and a fixed scaffold outer split, with target size approximately 20%. The outer seeds are 43021 and 43022, respectively. Target selection uses structure or scaffold identifiers without outcome stratification. Scaffold groups remain intact. The target counts are 1,451 for AMES, 122 for hERG, 95 for DILI, and 2,617 for CYP2D6 in each regime. Five fits with seeds 311–315 use the remaining source pool and the same internal fit/router/calibration allocation. Repeated fit seeds never redraw the target cohort. The extension is a within-dataset holdout across additional endpoint families; it is not an independently collected cohort or a blinded prospective study. In particular, the DILI target count is below the historical 100-target audit threshold, and the hERG and DILI internal calibration sets also fall below the historical 100-source threshold. Those settings remain fixed so that the small-sample fallback behavior is visible. CYP2D6 was added before its predictive comparison on the basis of documented sample size, to provide a second larger non-Tox21 family alongside AMES. It does not replace either small endpoint, and all four tasks are retained regardless of model performance or gate activation.

2.4 Conformal sets and the finite-sample rank

Let $p_y(x)$ denote the fitted probability for $y \in \{0,1\}$. The candidate-label LAC score is

$$s(x,y) = 1 - p_y(x). \tag{1}$$

For n calibration scores, define $k = \lceil (n+1)(1-\alpha) \rceil$. The threshold is the kth smallest score when $k \leq n$ and $+\infty$ otherwise. Empty calibration groups also receive $+\infty$. In particular, the rank is never truncated to n. Marginal calibration uses all observed calibration scores. Mondrian calibration uses only the scores

whose observed outcome equals the candidate label, giving thresholds q_0 and q_1 and set

$$S_M(x) = \{y : s(x,y) \leq q_y\}. \tag{2}$$

The candidate label, rather than the point-predicted class, selects its Mondrian threshold.

For deterministic APS, labels are ordered by decreasing fitted probability, with a fixed class-index tie order. The score for a candidate label is the cumulative probability through that label in this ordering. The same scoring rule is used on calibration and target rows. This deterministic version is compared as a component sensitivity analysis; it is not presented as a new score or as an exact reproduction of every randomized APS variant.

2.5 Estimated weighting and transport diagnostics

The domain-classifier baseline estimates a target-to-source density ratio from unlabeled source and target covariates. Posterior odds from logistic domain classification are multiplied by the source-to-target sample-size ratio and clipped at the specified bounds. The historical views comprise nine physicochemical descriptors, component and stack logits, and diagnostic ECFP bits. Source and target counts, domain AUC, class-specific and overall effective sample sizes, clipping frequencies, and weighted mean discrepancy are retained as diagnostics.

For calibration weights w_i and a target weight $w(x)$, the weighted calibration distribution for candidate class y is

$$F_{x,y} = \frac{\sum_{i:Y_i=y} w_i \delta_{s(X_i,Y_i)} + w(x)\delta_{+\infty}}{\sum_{i:Y_i=y} w_i + w(x)}. \tag{3}$$

The $(1-\alpha_y)$ quantile gives the corresponding threshold. The target mass at infinity is retained. With no class-y calibration observation, that candidate has an infinite threshold.

Likelihood-ratio conformal theory requires the relevant distributional assumptions and weights [2]. Here, ratios are estimated, clipped, and constructed in selected molecular views. A high effective sample size or a low domain AUC does not establish the correctness of those assumptions. Passing the historical gates permits the empirical correction to be used; it does not certify conditional coverage. An ungated weighting comparator and gate sensitivities are therefore necessary to evaluate the gates' empirical contribution.

2.6 Protected classes and screening actions

For Tox21, label 1 means assay-active and label 0 means assay-inactive according to the released outcome encoding. For the four additional families, these are positive and negative endpoint-specific labels with the meanings specified above. We map $\{0\}$ to an inactive decision, $\{1\}$ to an active decision, and the full set to deferral. Raw empty sets are reported separately. For operational expansion comparisons, an empty baseline is repaired to the full set for every method before additional deferral is counted.

The historical TAME protects label 0, chosen after observing a development coverage deficit. It can turn an erroneous active decision for an inactive compound into deferral, but cannot repair the active class by adding label 1. The revised active-protection question instead fixes label 1 from the stated action: avoiding an inactive decision for an assay-active compound. These are different objectives. Both variants, along with protection of both labels, are reported; a new active-protection result cannot be retroactively attributed to the original method.

Let S_0 be the repaired Mondrian baseline and W_v a weighted set from view v. Under the historical two-view consensus rule, a protected label is added when both eligible views contain it. If the required views are unavailable, the output remains S_0. Single-view, union, ungated, and protected-class variants isolate the role of each component. For every expansion variant S, the invariant $S_0(x) \subseteq S(x)$ is checked. It establishes nondecreasing empirical coverage on the same observations and nonincreasing singleton yield; it does not establish useful error targeting.

2.7 Matched deferral budgets

For a reference expansion rule, define the additional number of deferred baseline singletons of each predicted class as

$$m_c = \sum_i \mathbb{1}\{S_0(x_i) = \{c\},\ |S(x_i)| = 2\}, \qquad c \in \{0,1\}. \tag{4}$$

Each matched comparator expands exactly m_0 baseline class-0 singletons and m_1 baseline class-1 singletons. This matches both the total budget and the composition of the original decisions. It prevents an apparent benefit from arising simply because one rule defers more inactive decisions in an imbalanced assay.

The primary simple comparator ranks the probability assigned to the omitted label: among baseline $\{0\}$ decisions, it defers larger $p_1(x)$ first; among baseline $\{1\}$ decisions, it defers larger $1 - p_1(x)$ first. A

Mondrian singleton need not equal the probability argmax, so this decision-specific ranking need not agree with generic entropy or maximum-probability uncertainty. It directly targets the candidate label whose addition could repair the decision. The implementation identifies the active-protection matched comparator as `matched_label1_active_omitted_label_probability`.

Secondary rankings use lower maximum predicted probability, lower maximum training-set Tanimoto similarity, or fixed seeded random priorities. Stable molecule identifiers resolve ties. All rankings use predictions, structures, and fixed seeds only; target outcomes never rank compounds or select a budget. The reference TAME budget is itself computed from label-free outputs on the target batch; this is an explanatory same-budget comparison, not a claim that every rule achieves a prespecified clinical workload. Fixed deferral-rate curves provide a complementary view, including the zero-additional-deferral and always-defer limits.

2.8 When transport expansion reduces to probability ranking

Consider binary LAC, a fixed calibration sample, a fixed set of active views, and protection of label 1. If the query weight in each view v is a positive constant across all target compounds, its weighted class-1 threshold q_{1v} is also constant. Adding label 1 in that view is then equivalent to $p_1(x) \geq 1 - q_{1v}$. Across the active views, intersection uses $p_1(x) \geq \max_v(1 - q_{1v})$ and union uses $p_1(x) \geq \min_v(1 - q_{1v})$. Restricted to baseline $\{0\}$ decisions, both therefore select an upper level set of p_1. At the same expansion count, omitted-label-probability ranking selects the identical set. Ties do not alter this conclusion because a fixed threshold includes every member of a boundary tie, with a common deterministic ordering used for any budget selection. If the required views fail the gate, both methods have zero additional budget.

This observation is a restricted structural identity, not a validity theorem. Source reweighting and cell-level gates can change the threshold and therefore the workload, but under these conditions they cannot supply a different within-pool ranking. Additional ranking information must come from query-varying weights or from a rule outside this setting. We consequently evaluate each ungated, union, single-view, and threshold-sensitivity expansion against its own class-budget-matched probability rule. Comparing expansions with different deferral counts is retained as a workload description and is not interpreted as evidence that a gate improves error targeting.

2.9 Comparators and sensitivity analyses

The controlled suite includes marginal LAC, Mondrian LAC, Mondrian APS, ordinary Mondrian with empty-set repair, domain-ratio-weighted Mondrian for each view, the historical TAME, the active-label and both-label variants, and matched omitted-label-probability, uncertainty, applicability-domain, and random deferral. The ordinary equal-logit predictor is retained in the base-model comparison. Alpha and deferral-budget curves are reported rather than selecting an operating point after inspecting EPA outcomes.

The end-to-end reference suite contains independent paper-based implementations of CoDrug and KMM-CP. These implementations are distinguished from author-code execution and from the optional density-estimator components. Both use AttentiveFP backbones, with their respective training settings, while the common classical predictors isolate post-prediction effects. Consequently, the end-to-end comparisons measure the complete systems and are not attributed solely to the uncertainty wrapper.

CoDrug combines classification loss with an energy-margin penalty, uses an auxiliary ZINC molecular pool filtered by source scaffolds, and estimates separate calibration and target densities in the model's logit space. The implementation uses a 512-dimensional graph representation, classifier hidden sizes 256 and 8, two message-passing layers, two readout iterations, Adam learning rate 0.001, batch size 64, at most 100 epochs, and router-role cross-entropy early stopping with patience 15. Gaussian-KDE bandwidths are selected separately by ten-fold cross-validation over twenty-five fixed candidates between $10^{-1.3}$ and 10. Weighted global and candidate-label Mondrian sets retain the query mass at infinity. The global construction follows the published CoDrug equations; the candidate-label Mondrian version is an explicitly named extension for the present class-specific comparison. The printed energy margins $m_{\text{in}} = -5$ and $m_{\text{out}} = -35$ are evaluated as written. Because their ordering may conflict with the intended in/out energy separation, a second predeclared implementation swaps them to -35 and -5. Both versions are reported; the more favorable or less favorable version is not selected after inspecting outcomes. The energy weight is 0.01 and the temperature is one.

KMM-CP uses an independently trained AttentiveFP backbone without CoDrug's energy penalty or auxiliary molecules. It uses a 512-dimensional graph representation, one 64-dimensional classifier hidden layer, three message-passing layers, inverse-frequency cross-entropy weighting, Adam learning rate 0.001, a per-epoch learning-rate factor 0.8, batch size 128, at most 50 epochs, and training-loss patience five. The 64-dimensional penultimate representation is supplied to kernel mean matching. Five RBF

bandwidth multipliers, 0.01, 0.1, 0.5, 1, and 2 times the cross-domain median distance, are compared using a permutation-standardized unbiased squared MMD with 100 fixed-seed permutations. Bounded KMM uses weight bound 30 and an exact mean-mass constraint. Selective KMM jointly estimates source weights and target selection weights, requires mean selection mass at least 0.5, retains target weights at least 0.2, and then refits ordinary KMM to the retained targets. Solver status and constraints are checked; a failed optimization does not silently become an ordinary conformal result.

The published KMM-CP pseudocode applies the predicted class's threshold to every candidate, whereas its class-conditional description implies a threshold selected by the candidate label. We therefore report both interpretations explicitly, together with the global version, for ordinary and selective KMM. Their empirical weighted quantiles have no query mass at infinity or $(n+1)$ rank correction, following the published procedure. No finite-sample likelihood-ratio guarantee is assigned to this construction. Exact mass tolerance, permutation settings, and other underspecified choices are disclosed as implementation decisions. For selective methods, unselected compounds defer in the common full-cohort analysis; accepted-only coverage is also reported with its own denominator.

Optional Gaussian-KDE ratio and fixed-kernel mean-matching components provide additional estimator diagnostics. The KDE component uses the target mass at infinity in the weighted quantile above. The kernel-mean-matching component uses a weighted empirical quantile without a target mass; this distinction is recorded explicitly, and no finite-sample weighted-conformal guarantee is assigned to that component. Neither diagnostic includes all training, bandwidth-selection, and support-selection choices of the corresponding published end-to-end method.

Sensitivity analyses vary one historical gate at a time, remove all gates, evaluate each view alone, change the view quorum, and change the protected-label set. The original numerical thresholds are treated as study-specific settings. Their effect is assessed by activation frequency, error targeting, and deferral cost across assays; no threshold is called transferable solely because it passes on this collection. The public leaderboard can support exploratory configuration work, while the retrospective EPA comparison uses the recorded configuration without further outcome-driven retuning.

To examine query-specific transport information, five fixed-seed controls permute query density-ratio weights among molecules within fixed probability bins $[0,0.2),\ldots,[0.8,1]$. The same permutation is applied across both views, retaining their dependence. Source calibration weights, the target-density fit,

and the joint view-weight distribution remain unchanged. Both gated and ungated controls are evaluated, and each receives its own exact class-matched omitted-label-probability comparator because permutation can change its deferral count. This is a query-weight alignment diagnostic; it is not a destruction of the full target distribution or an independent replication of the study.

2.10 Controlled shift mechanisms

Four synthetic settings separate density information from the conditional outcome mechanism. The fixed source posterior is $p_1(x) = \text{logit}^{-1}(2x_1 + 0.5x_2 - 1)$; it is not refitted in response to target labels. The two views use the raw two-dimensional covariates and their elementwise squares. Thirty fixed-seed replicates each contain 400 calibration and 800 target observations at alpha 0.10. The no-shift setting uses standard Gaussian covariates. The overlapping covariate-shift setting moves the target mean by one in its first coordinate and retains the source conditional outcome law. Its exact, unclipped marginal density ratio is $\exp(x_1 - 1/2)$, supplying a known-weight comparator.

The support-loss setting places half the targets outside the bounded calibration support, where the required density ratio is unavailable. Finally, a changed-conditional setting reuses exactly the no-shift calibration covariates, labels, target covariates, probabilities, and transport diagnostics, while replacing the target posterior by $1 - p_1(x)$. This construction directly tests a limitation of label-free diagnostics: identical observable inputs cannot certify that the conditional outcome law is unchanged. The simulations are mechanism checks and are not pooled with molecular evidence. No setting supplies a universal weakest-class guarantee for a rule that protects only label 1.

2.11 Outcomes and paired inference

For each true class y, coverage and correct-singleton yield are

$$C_y = \Pr\{y \in S(X) \mid Y = y\}, \tag{5}$$

$$\text{CSY}_y = \Pr\{S(X) = \{y\} \mid Y = y\}. \tag{6}$$

We report both classes, $\min(C_0, C_1)$, and $\text{MacroCSY} = (\text{CSY}_0 + \text{CSY}_1)/2$. Decision-specific outcomes are false-inactive exposure $\Pr\{S(X) = \{0\} \mid Y = 1\}$, false-active exposure $\Pr\{S(X) = \{1\} \mid Y = 0\}$, the active

fraction among inactive decisions $\Pr\{Y = 1 \mid S(X) = \{0\}\}$, inactive-decision yield, total singleton rate, deferral rate, and raw empty-set rate. Operational deferral and singleton counts use every eligible target, including compounds with unobserved outcomes. Coverage and error metrics use observed outcomes only; labeled-population deferral is retained separately. A missing class or an empty decision denominator produces an undefined value with its count; it is not silently treated as zero.

The primary contrast uses the original stack, LAC scores, $\alpha_0 = \alpha_1 = 0.10$, and the historical six-assay panel. It compares active-protection TAME against the class-matched omitted-label-probability rule on false-inactive exposure, with identical eligible rows and additional deferral budgets. A negative TAME-minus-comparator difference favors TAME. Uncertainty, applicability-domain, and random matching are secondary comparisons; the all-twelve-assay panel and alternative predictors are sensitivity analyses.

Each metric is computed separately for each assay–seed cell, and the cell metrics are averaged with equal weight across the declared assays and seeds. Probabilities or set memberships are not averaged across seeds before thresholding. Repeated seeds do not multiply the number of independent compounds. Paired bootstrap resampling uses molecule or standardized-structure identity as a shared resampling unit across assays and methods, retaining all seed predictions for each selected unit and recomputing the same cell metrics and equal-weight summary. This preserves cross-assay dependence induced by repeated compounds. Intervals describe sampling variation for the evaluated collection and fixed assay panel, rather than generalization to all molecular assays.

An additional descriptive analysis identifies molecules deferred only by TAME and only by its matched probability comparator. It counts active and inactive outcomes in these discordant sets and stratifies them by fixed probability and training-similarity bins. Budget equality is verified on all eligible target structures. If some outcomes are missing, the labeled subsets can have different expansion counts; the error–correct-singleton decomposition retains that observed-budget correction. These counts explain a measured difference without treating every deferred molecule as a recovered error.

No five-percentage-point efficiency tolerance is interpreted as a clinically validated utility threshold. Error–deferral curves and their paired contrasts replace a binary promotion claim. An advantage in coverage against a less-deferred baseline alone is insufficient evidence of a useful expansion rule. Whether the revised study supports TAME, a simpler comparator, or no stable winner will be determined from the complete comparisons, including unfavorable findings.

3 Results

3.1 Archived findings that motivate the revision

The original six-assay comparison contained thirty assay–seed cells. Relative to ordinary stacked Mondrian prediction, historical TAME increased mean minimum-class coverage by 1.3649 percentage points and decreased MacroCSY by 1.6067 points. Its archived hierarchical-bootstrap interval for the coverage contrast was 0.5827 to 2.0051 points. This is the original analysis, not a newly estimated interval under the revised cross-assay molecule-resampling procedure. The class-1 coverage change was zero in every reported primary assay because the intervention protected label 0.

The archived twelve-assay summary provides an additional comparison that must not be omitted (Table 1). Equal-logit ordinary Mondrian had slightly higher mean minimum-class coverage and MacroCSY, and a lower ambiguous-set rate, than TAME. These descriptive averages do not establish statistical superiority, and they are not substitutes for a paired six-assay analysis. They do establish why the old TAME-versus-stacked-Mondrian contrast cannot support a general superiority claim.

Table 1: Archived descriptive means across twelve assays and five seeds. All values are percentages. These are historical results, not outputs from the revised experiment suite.

Method	Minimum coverage	MacroCSY	Ambiguous
Ordinary Mondrian, stacked	83.35	42.38	49.05
Ordinary Mondrian, equal-logit	84.41	41.44	49.57
Historical TAME, label 0	84.30	41.33	50.40
Weighted Mondrian, physicochemical	83.60	43.09	46.98
Weighted Mondrian, score view	82.48	42.52	45.95
Weighted Mondrian, ECFP	99.80	0.22	99.65

3.2 Class-specific error at matched deferral

Experiment pending: Report the primary active-protection comparison against matched omitted-label-probability deferral under the stacked predictor at alpha 0.10 in the six-assay panel. Then report inactive-protection, both-label, uncertainty, similarity, and random comparisons as declared secondary analyses. Include exact per-class expansion counts, point effects, paired intervals, and assay-level results. Do not infer error-targeting value from set inclusion alone.

3.3 Strong conformal comparators and model sensitivity

Experiment pending: Complete fixed-predictor LAC/APS and weighting comparisons, the equal-logit and logistic predictor analyses, and externally verified CoDrug and KMM-CP comparisons. Distinguish component implementations from full methods and show all declared operating-point curves.

3.4 Contribution of transport views and gates

Experiment pending: Quantify single-view, quorum, gate-removal, gate-threshold, and protected-class effects. Report how often a method falls back or defers nearly every compound. Include configurations whose coverage gain is explained entirely by deferral volume.

4 Discussion

The central question is whether an uncertainty rule chooses the right compounds for deferral at a specified workload. This differs from asking whether adding labels improves empirical coverage: any set expansion satisfies the latter weak inequality. The matched-budget design tests the information contributed by transport views after controlling that mechanical effect and the original predicted-class composition.

The distinction between active and inactive decisions also changes the interpretation of the historical method. Adding label 0 protects inactive compounds from an erroneous active singleton. Adding label 1 protects active compounds from an erroneous inactive singleton. Neither is a universal definition of the costly error. A screening study must state its action and assay meaning first; a development-selected coverage deficit cannot substitute for that statement. The Tox21 analysis treats assay activity as the measured outcome and does not translate it into patient safety or clinical toxicity.

The archived equal-logit result is a useful check on methodological complexity. It shows that an apparently favorable contrast against one stacked predictor does not establish that a more elaborate wrapper improves the attainable coverage–informativeness trade-off. The revised design accordingly retains simple predictors and simple deferral rules alongside shift-aware conformal methods. If a simpler rule matches or improves upon TAME at the same budget, that is an informative result and should change the conclusion.

Transport diagnostics may still be useful even when they do not yield a new validity result. They reveal weight concentration, poor overlap, and sensitivity to the representation used to estimate shift. Their

decision value, however, requires an empirical comparison with ungated and alternative rules. A failed gate disables an operation; a passed gate does not prove that the operation is statistically justified. Threshold sensitivity and external assay evaluation are needed before adopting the same numerical settings elsewhere.

Multiple assays share compounds and are not independent replications of a universal molecular prediction task. Repeated seeds describe variability from training partitions, not additional biological samples. The revised inference retains this dependence and reports sparse denominators explicitly. Because the EPA outcomes are already known, this revision can supply a transparent retrospective comparison but cannot claim a second independent validation on that collection. Additional assay-family evaluations are reported separately with their label semantics and partition design; they are not combined with the Tox21 primary panel to enlarge its denominator.

Experiment pending: Replace this paragraph with a result-specific conclusion only after the declared comparison matrix and sensitivity analyses are complete. A positive conclusion requires evidence of useful class-specific error targeting beyond matched simple rules; otherwise the paper should conclude that the evaluated transport expansion does not show that advantage and present the resulting chemometric comparison without an algorithm-superiority claim.

Data and code availability

The historical source and revised experiment suite are maintained in the project repository. The revision keeps the submitted analysis separate from new outputs. Source acquisition, shared-row comparisons, row-level prediction export, tests, and execution instructions are provided with the code. Completion status and experiment provenance must accompany every new result table.

Author declarations: revision status

The original affiliations and author order are retained in this working draft. Both authors must review the completed analyses, update their actual contributions, and approve the final manuscript before submission. Funding, competing-interest, and AI-assistance declarations must be checked at that time. This draft does not certify author approval or numerical verification that has not yet occurred.

References

[1] V. Vovk, Conditional validity of inductive conformal predictors, Proceedings of Machine Learning Research 25 (2012) 475–490. https://proceedings.mlr.press/v25/vovk12.html.

[2] R. J. Tibshirani, R. F. Barber, E. J. Candès, A. Ramdas, Conformal prediction under covariate shift, Advances in Neural Information Processing Systems 32 (2019). https://proceedings.neurips.cc/paper/2019/hash/8fb21ee7a2207526da55a679f0332de2-Abstract.html.

[3] S. Laghuvarapu, Z. Lin, J. Sun, CoDrug: Conformal drug property prediction with density estimation under covariate shift, Advances in Neural Information Processing Systems 36 (2023). https://proceedings.neurips.cc/paper_files/paper/2023/hash/7691484a7a35d5e2742279c1d926b778-Abstract-Conference.html.

[4] S. Laghuvarapu, R. Deb, J. Sun, KMM-CP: Practical conformal prediction under covariate shift via selective kernel mean matching, Proceedings of Machine Learning Research 337 (2026) 3237–3261. https://proceedings.mlr.press/v337/laghuvarapu26a.html.

[5] Y. Romano, M. Sesia, E. Candès, Classification with valid and adaptive coverage, Advances in Neural Information Processing Systems 33 (2020). https://proceedings.neurips.cc/paper_files/paper/2020/hash/244edd7e85dc81602b7615cd705545f5-Abstract.html.

[6] R. Huang, M. Xia, D.-T. Nguyen, T. Zhao, S. Sakamuru, J. Zhao, S. A. Shahane, A. Rossoshek, A. Simeonov, Tox21Challenge to build predictive models of nuclear receptor and stress response pathways as mediated by exposure to environmental chemicals and drugs, Frontiers in Environmental Science 3 (2016) 85. https://doi.org/10.3389/fenvs.2015.00085.

[7] C. Xu et al., In silico prediction of chemical Ames mutagenicity, Journal of Chemical Information and Modeling 52 (2012) 2840–2847. https://doi.org/10.1021/ci300400a.

[8] S. Wang et al., ADMET evaluation in drug discovery. 16. Predicting hERG blockers by combining multiple pharmacophores and machine learning approaches, Molecular Pharmaceutics 13 (2016) 2855–2866. https://doi.org/10.1021/acs.molpharmaceut.6b00471.

[9] Y. Xu et al., Deep learning for drug-induced liver injury, Journal of Chemical Information and Modeling 55 (2015) 2085–2093. https://doi.org/10.1021/acs.jcim.5b00238.

[10] H. Veith et al., Comprehensive characterization of cytochrome P450 isozyme selectivity across chemical libraries, Nature Biotechnology 27 (2009) 1050–1055. https://doi.org/10.1038/nbt.1581.